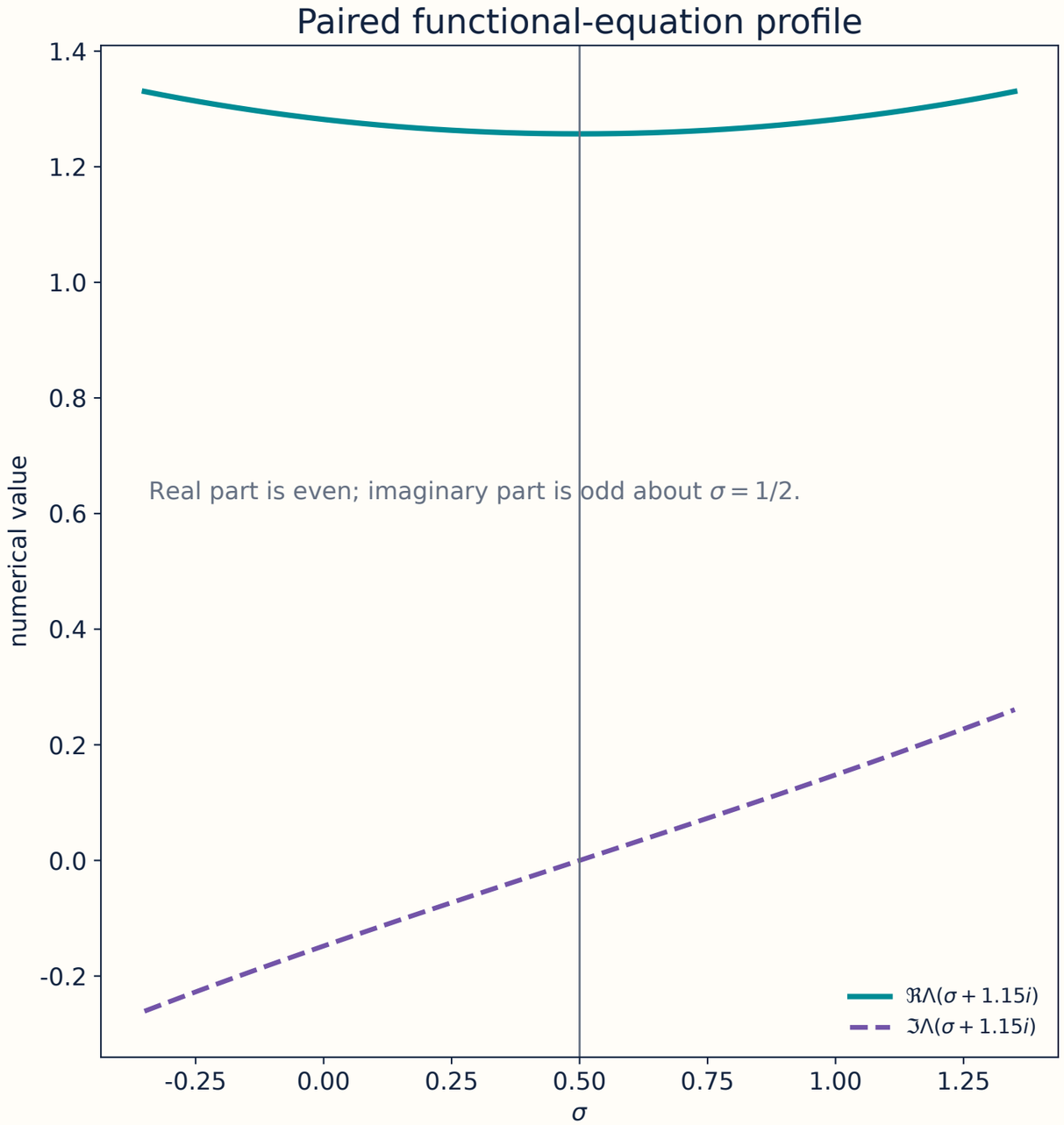
Gaussian periodic sum → theta transformation → Mellin integral → completed series

$$\Lambda(s) = (5/\pi)^{(s+1)/2}\Gamma((s+1)/2)\,5^{-s}\sum_{j=1}^{4} a_j \zeta(s, j/5)$$

For the Fourier eigenvector: $\Lambda(s) = \Lambda(1-s)$